

- Q.31 What is Stub? Explain the principal of impedance matching using stub.
- Q.32 A lossless line is terminated in pure resistance of 600Ω if characteristic impedance is 400Ω . Find value of SWR.
- Q.33 Explain the concept of transmission line at high frequency.
- Q.34 Write a short note on Crystal filter.
- Q.35 List different types of transmission line.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is a symmetrical network? Explain concept of propagation constant and attenuation constant in a symmetrical 'T' network.
- Q.37 Write short note on
- Band Pass Filter
 - SWR and VSWR
- Q.38 What are the various type of distortion in transmission line and also explain the condition for minimum distortion.

(1160)

(4) 181035/171041/121041
/031041

No. of Printed Pages : 4 181035/171041/121041
Roll No. /031041

3rd Sem / Eltx, Power Eltx Subject:- Network Filters and Transmission Lines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 How many terminals a two port network has
- 1
 - 2
 - 3
 - 4
- Q.2 A two port is reciprocal or bilateral provided
- $Z_{11} = Z_{22}$
 - $Z_{11} = Z_{12}$
 - $Z_{12} = Z_{21}$
 - $Z_{21} = Z_{22}$
- Q.3 Characteristics impedance for asymmetrical network at both the ports is
- Same
 - Different
 - Defined
 - Not defined
- Q.4 In symmetrical T network what is the values of series arm impedance.
- Z_1
 - $Z_1/2$
 - Z_2
 - $Z_2/2$
- Q.5 For a prototype LPF, the phase constant β in the attenuation band is given by
- π
 - 0
 - ∞
 - $\pi/2$

(1) 181035/171041/121041
/031041

- Q.6 A band pass filter may be obtained using a high pass filter followed by
- Low Pass Filter
 - High Pass Filter
 - Band Pass Filter
 - Band Reject Filter
- Q.7 For a prototype LPF, series arm component is
- Capacitance
 - Inductance
 - Resistance
 - none of the above
- Q.8 Input impedance of short circuited loss line with length $\lambda/4$ is
- Infinity
 - Zero
 - Characteristic impedance
 - None of the above
- Q.9 An active network has
- an emf source
 - a current source
 - Both
 - None of the above
- Q.10 If $K=0$, Then VSWR will be _____
- 0
 - 1
 - 2
 - All of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 A two port network has _____ terminals.(two/four)
- Q.12 In an asymmetrical T-network, series arm in both sides have same impedance.(True/False)
- Q.13 Define attenuators.
- Q.14 Decibel is a unit of _____ (capacitance /attenuation)

(2) 181035/171041/121041
/031041

- Q.15 Inductor is an _____ element.(active/passive)
- Q.16 Name primary constant of transmission line.
- Q.17 Expand VSWR.
- Q.18 Write full form of LPF.
- Q.19 R,L,F and G are known as secondary constant of transmission line. (True/False)
- Q.20 $Z_{oc} * Z_{sc} =$ _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Derive an expression for characteristic impedance of 'T' network.
- Q.22 What is asymmetrical network? Explain iterative impedance of an asymmetrical network.
- Q.23 Differentiate between Linear and Non- linear network.
- Q.24 Explain concept and significance of Iterative Impedance.
- Q.25 Design a symmetrical 'T' attenuator.
- Q.26 What are the uses of attenuator?
- Q.27 Draw and Drive the design equation of L-type attenuator.
- Q.28 Draw and Explain the characteristic (phase shift v/s frequency) of T filter.
- Q.29 Explain concept of band pass filter.
- Q.30 What is difference between active and passive filter? Explain in detail.

(3) 181035/171041/121041
/031041